Measurement protocol V7.3

Section 13 · 12 November 2026 · one page

Every number comes from the JSON files in `dati/`: no value is written by hand.

Reference: R = 0.5 μm, λ = 800 nm, z = 2.0 μm, plasma 1.3 mPa · s, n = 1.33, T = 310 K.

Goal: measure latch, thermal, and gate; do not prove the engine.

Setup 1 — Latch: ratio and paired noise

Measurement. 3 fast half-turns against 3 slow half-turns on the same bead (r = 0.3969 μm, linker 1.00 μm) at the same power. A half-turn is half of the 720° cycle, i.e. one 360° turn. The fast half-turn (counterweight disengaged) is 0.5996 s, the slow one (counterweight engaged) 2.327 s; the sum is 2.926 s, the ratio is 3.881, up to 4.492 with the unvalidated hydrodynamic correction.

Expected. Ratio 3.881 for the ideal geometry; 4.256 with the translational bead term, 4.335 with the full term, 4.492 as the upper bound.

Error. CV = 17.84 %; σ_R/R = CV·$\sqrt{2/3}$ = 14.57 %; per-turn $\varepsilon = 1.07 \times 10^{-4}$ (geometric threshold) and 1.01×10^{-4} (optimized threshold); ε on sums of 3 = 1.95×10^{-11} (geometric).

Falsification. Ratio outside 3.881–4.492 by $2\sigma_R$ (below 2: no CW/No asymmetry); a measured CV above 17.76 % makes the 1×10^{-4} budget unreachable per turn, and above 30.77 % makes it unreachable on sums.

Note. The per-turn criterion does not improve with N; the sums improve with the square root of N. At CV = 30.2 % the per-turn error is 2.21×10^{-2} (outside budget) and the sums give 7.38×10^{-5} (inside). The total-duration test would give 1.14×10^{-2} and is not used.

Setup 2 — Thermal: sink-mirror bracket

Measurement. Steady-state ΔT in blood, R = 0.5 μm, z = 2.0 μm, at 800 nm and 1064 nm; incident power 0.6628 mW and 0.8507 mW.

Expected. At 800 nm: sink 0.3471 K, no correction 0.5086 K, mirror 0.6701 K. At 1064 nm: 0.2827 K, 0.3906 K, 0.4986 K.

Error. Model bracket ±0.161 K at 800 nm and ±0.108 K at 1064 nm; declared resolution 0.05 K; the criterion is 5.0 K.

Falsification. $\Delta T \geq 5.0$ K at either wavelength, or ΔT above the mirror by 2σ. The 5.0 K criterion is chosen, not a biological threshold.

Margin. The mirror stays below 5.0 K by a factor 7.46 at 800 nm and 10.03 at 1064 nm.

Setup 3 — Gate: shear threshold and escape

Measurement. Shear ramp in plasma (μ = 1.3 mPa · s–1.5 mPa · s) and in blood (μ = 8.0 mPa · s, σ_y = 10.0 mPa); escape at Q = 0.1.

Expected. In plasma $\gamma^\star = 2.700\,\mathrm{s}^{-1}$, with $2.340\,\mathrm{s}^{-1}$ at μ = 1.5 mPa · s; in blood $0.2777\,\mathrm{s}^{-1}$ with $M_y = 4\pi R^3 \sigma_y$, and band $0.1037\,\mathrm{s}^{-1}$–$0.4094\,\mathrm{s}^{-1}$. At Q = 0.1: $U_0 = 8.59\,kT$, $\tau_{\mathrm{esc}} = 200.7$ s.

Error. In plasma $\pm 0.180\,\mathrm{s}^{-1}$ (6.67 %), entirely from the μ band; in blood the band $0.1037\,\mathrm{s}^{-1}$–$0.4094\,\mathrm{s}^{-1}$ covers μ = 7.0 mPa · s–9.0 mPa · s and σ_y = 5.0 mPa–20.0 mPa. The escape convention is $\tau_D = \xi/k_{\mathrm{rad}} = 0.0374$ s, not $w_0^2/(4\mathrm{D}) = 0.321$ s.

Falsification. Persistent rotation at $267.0\,\mathrm{s}^{-1}$ (venule, 98.88× the threshold) or $\gamma^\star$ outside the band by 2σ; no escape within 2007 s at Q = 0.1 falsifies the barrier.

Note. The Bingham prefactor is $4\pi R^3 \sigma_y$, conservative; the exact sphere solution would be a factor 1.5 lower. Whole blood remains an effective rheology, not a measured curve.

Common criteria

- **Threshold before data.** The geometric threshold is fixed before looking at the times; the optimized one is reported only as a check.
- **Pairing.** Every fast cycle is paired with its slow cycle; the total-duration test is not used.
- **Budget.** The budget is 1×10^{-4} per comparison; the maximum CV is 17.76 % per turn and 30.77 % on sums.
- **Report.** Value, error, and falsification are reported together; an out-of-band result is declared, not discarded.

What this protocol does not prove

- **It does not prove biological safety.** 5.0 K is a chosen thermal criterion, not a damage threshold.
- **It does not prove the latch.** It measures ratio and noise; it does not measure actuation, energy, or fatigue.
- **It does not prove operation in flow.** It measures the stasis threshold; whole blood remains an effective rheology, not a measured curve.